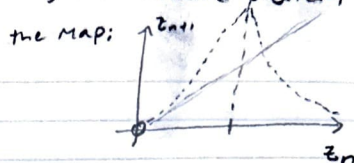
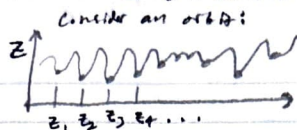
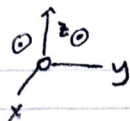


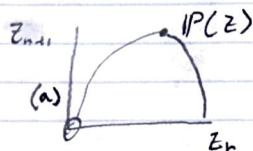
Lecture 10-30: 1-Dimensional Maps

→ consider a system in 3-Dimensions such that it possesses an attractor that is 2-Dimensional: i.e., the Lorenz system

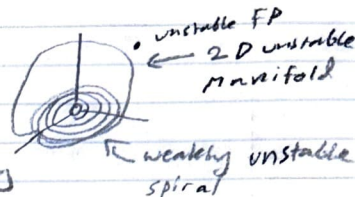


→ All neighborhoods of the Lorenz system are unstable to trajectories

→ Consider the Bössler system:



← exhibits a "stretching and folding" (like a toffee machine)



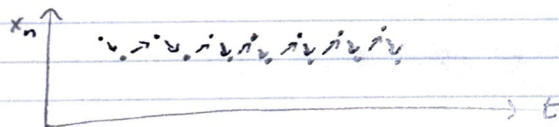
→ locally, trajectories get stretched out and folded over

→ globally, they get ^{*}very common example of how unstable flow maintain consistency with a bounded state space in 3+ Dimensions

→ Trajectories that are very far from each other can wind up in the same neighborhood

→ The map in (a) is actually a fractal (many close, repeated curves): We approximate it as a single curve, that is,

→ $f(x_n) = x_{n+1} \rightarrow x_{n+1} = r x_n (1 - x_n)$ (throw away Bössler, consider the map)



→ IF we have a quadratic mapping, $x_{n+1} = r x_n (1 - x_n)$

$$\Rightarrow x_2 = r x_1 (1 - x_1) = a x_0^2 + b x_0^3 + c x_0^4 + d x_0^5$$

$$\Rightarrow x_3 \propto O(x_0^6)$$

→ each step, the logistic map grows in 2 orders

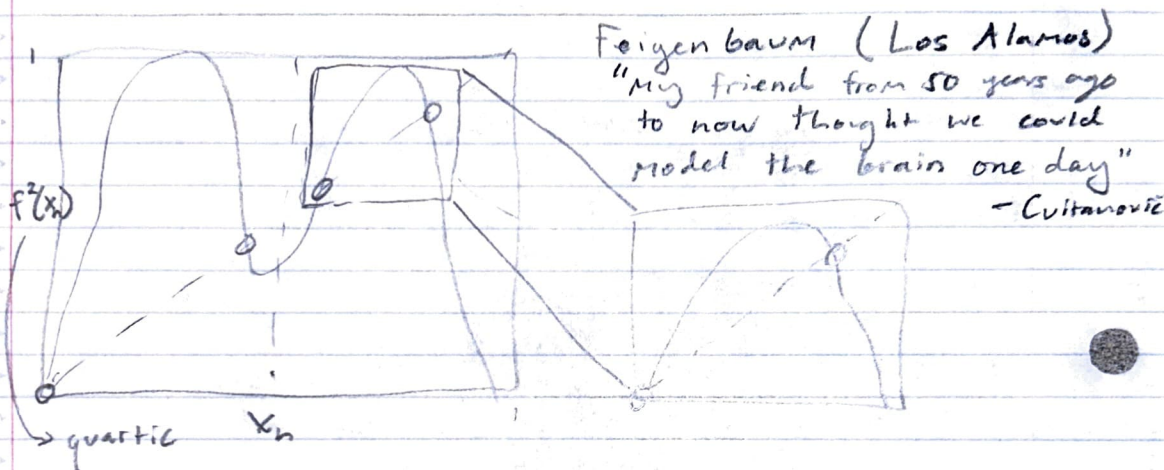
• not tractable to compute high periods $\gg x_1$

• keep in mind, the parabola is a simple way to understand the geometry of a complex, higher dimensional flow

Strogatz: Example 10.3.2

→ Consider the iterated map f :

$$\begin{aligned}x_{n+2} &= f(x_{n+1}) = f \circ f(x_n) \\ \Rightarrow x_{n+3} &= f \circ f \circ f(x_n) = f^3(x_n)\end{aligned}$$



→ Consider an iterated map:

- take a subset of the x_{n+1} iterator function map,
size of $\alpha = 2, 50, \dots$, $f(x) \approx \alpha f(\frac{x}{\alpha})$
- the solution of the map in that subset reflects the 'global' solution! Qualitative & quantitative
- Allows you to describe the long term evolution of the map